Syracuse University
School of Information Studies

Week 11
Building a Shiny App

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What is Shiny?

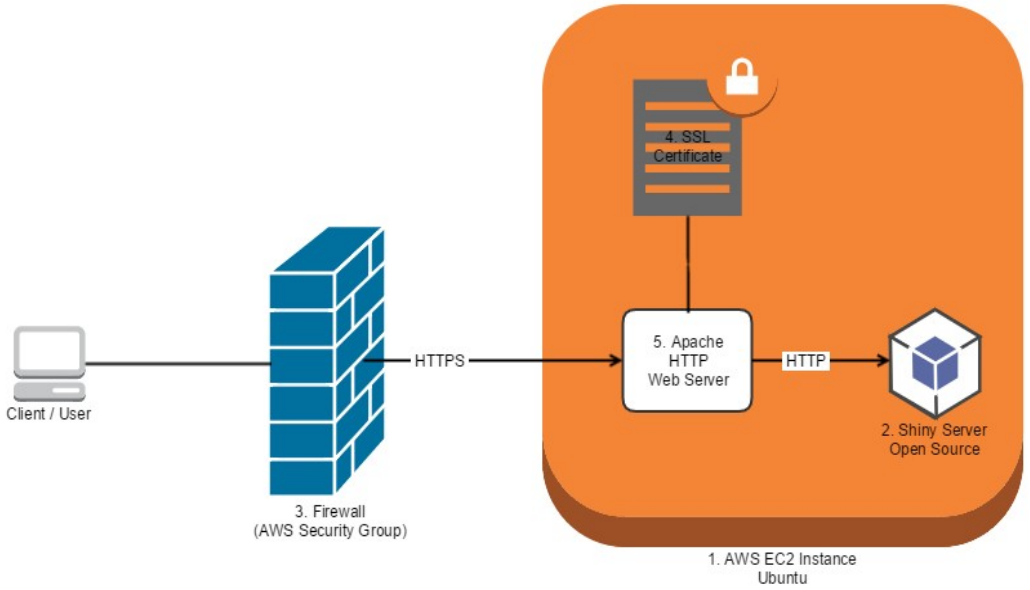
- This picture shows how a user's computer securely connects to an application running on a server in the cloud.
- Here is a breakdown of the steps and components in simple terms:

The Secure Connection

- **Client / User (Your Computer):** This is you, sitting at your computer, wanting to see a website or use an application.

- **Firewall:** This acts as the first line of defense. It checks all incoming and outgoing messages and makes sure only safe, expected traffic (like the HTTPS tunnel) is allowed to reach the main computer.

- **HTTPS:** When you type https:// in your browser, it means the connection between you and the server is encrypted.



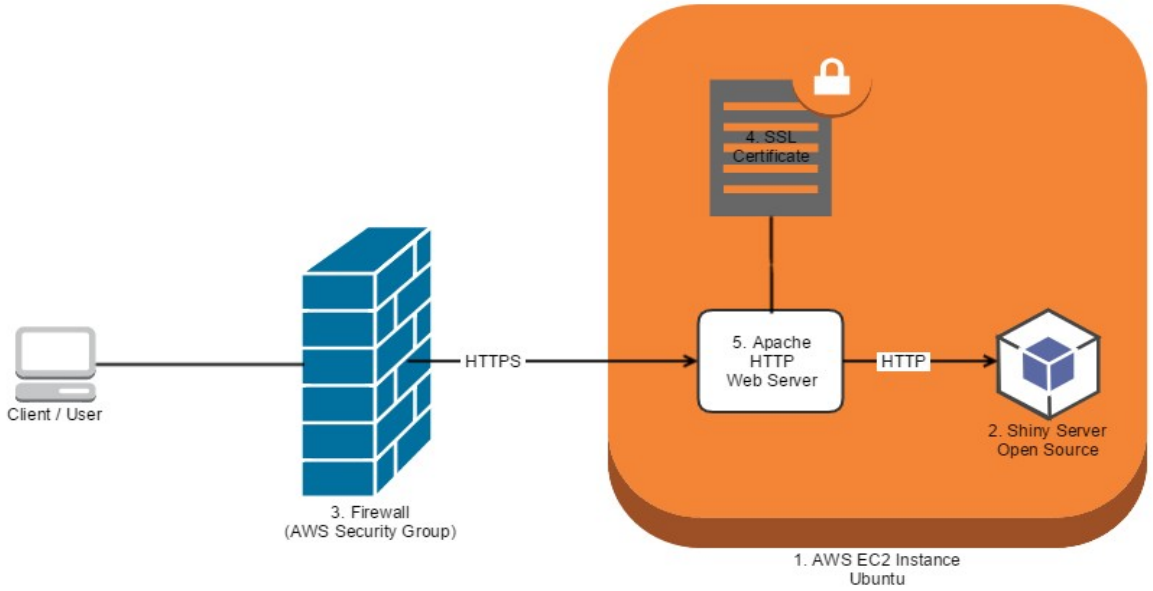
Inside the Server:

- The orange box is the **server**, where the application lives.

- **SSL Certificate (The Key):** It's the essential "key" that unlocks the **HTTPS** tunnel.

- **Apache HTTP Web Server:**
- This is the main piece of software that listens for requests from the internet.
- When your secure request (HTTPS) arrives, Apache uses the SSL Certificate to decrypt it.

- **Shiny Server Open Source (The App):** This is the actual application the user wants to run.



- AWS stands for **Amazon Web Services**.
- Think of it like a giant company (Amazon) that owns massive warehouses full of powerful computers and hard drives.
- Instead of keeping all those computers and hard drives for themselves, they rent out the power and storage to other people and companies so they can run their apps and websites.
- When you use AWS, you aren't using a physical computer sitting next to you; you're using a computer over the internet, in the "cloud.“
- **Renting Power:** Instead of buying your own expensive servers, you can rent just the amount of computing power, storage, and other services you need from AWS.
- This is where the term **"EC2 Instance"** comes from in the picture, it's like renting a specific computer (an **Elastic Compute Cloud** instance) just for your application to run on.
- **The Guard:** The **"AWS Security Group"** (the Firewall) in the image is a tool offered by AWS to help you manage the guard for your rented computer.

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The End